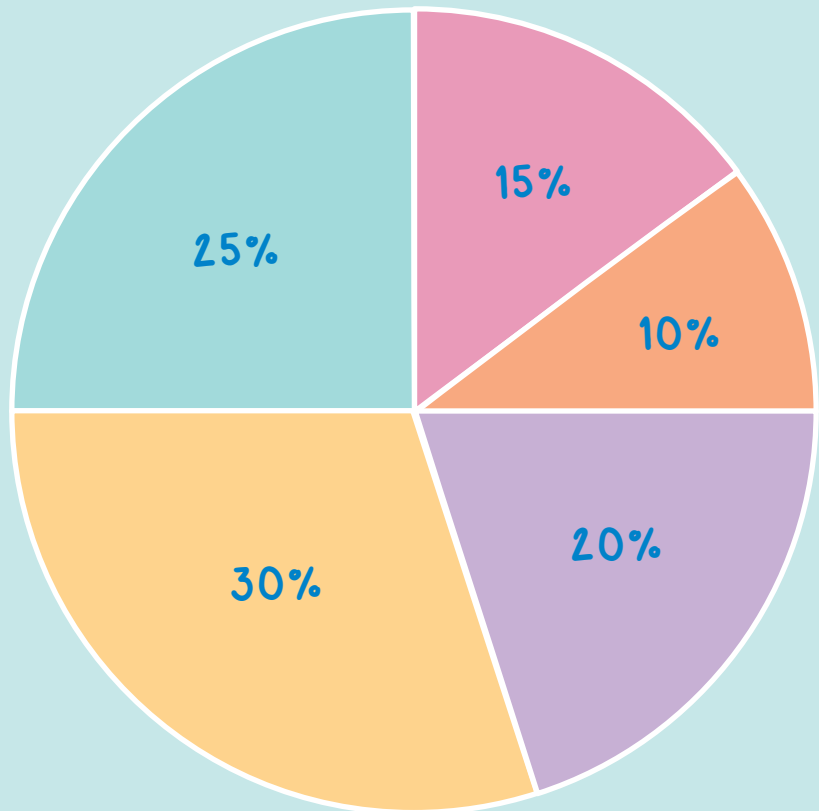
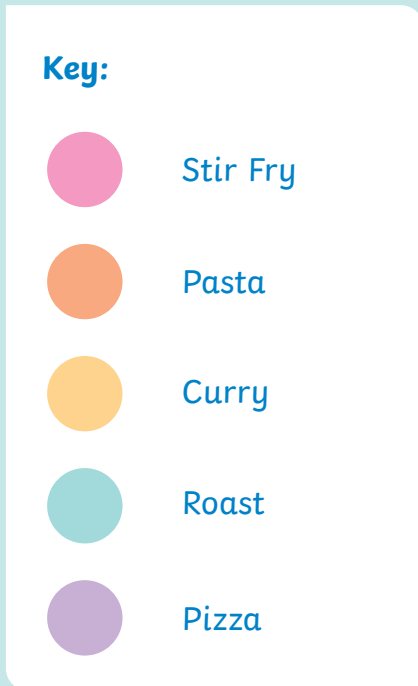


Statistics: Read and Interpret Pie Charts

Finding Amounts from Percentages

If we have a pie chart with percentages on, we can find out how much each segment is worth if we know the overall total. Here is an example:

Children's Favourite Food



Imagine that 80 children have voted on their favourite food. We can now work out the amount of votes each food received by working out the percentages of 80:

10% of 80 = 8 and we can use this to work out the other percentages:

Pasta was 10%, so 8 children chose pasta.

Pizza was 20%, so we can double the 10% value to find the total: $2 \times 8 = 16$ children.

Curry was 30%, so we can multiply the 10% value by 3 to find the total: $3 \times 8 = 24$ children.

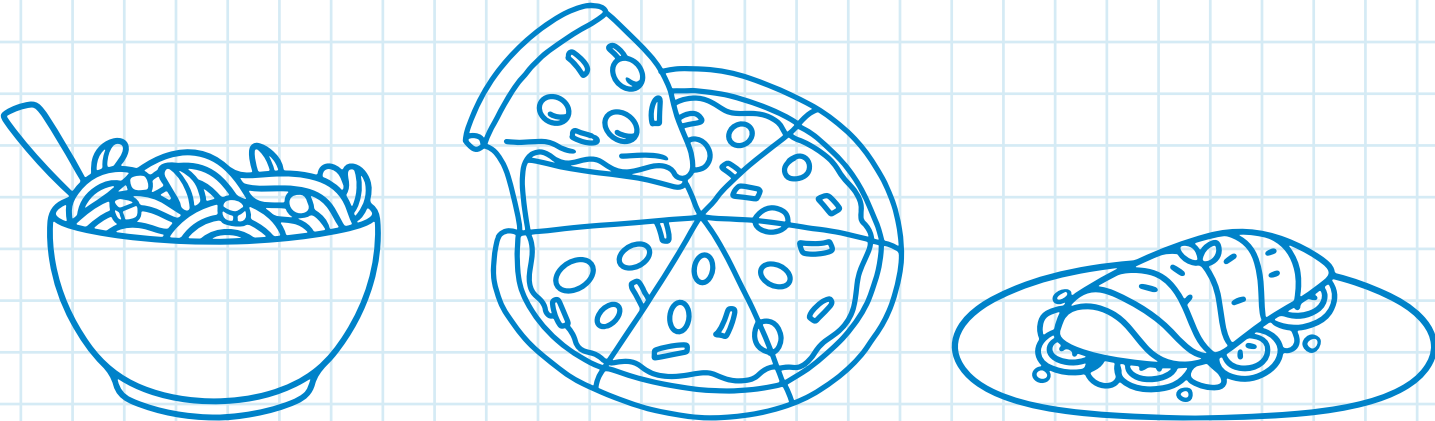
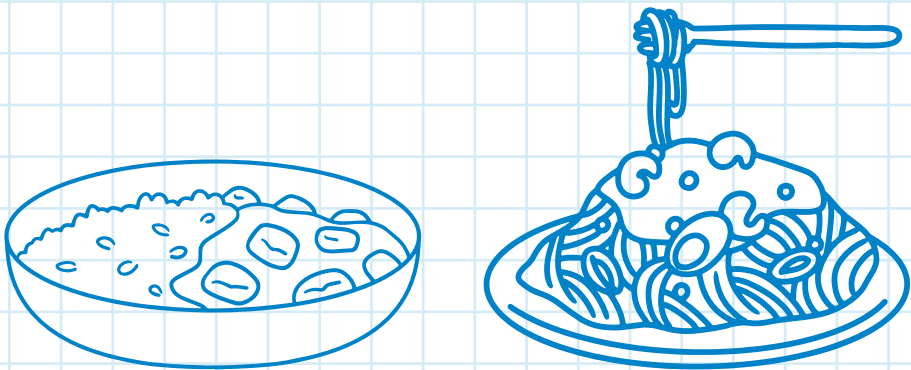
Roast was 25%, which is the same as $\frac{1}{4}$ of 80 which is 20.

15 children chose stir fry.

We can halve the 30% amount to find this total: $24 \div 2 = 12$ children.

Let's put our data into a table:

Colour	Category	Percentage	Number of Children
	Pasta	10%	8
	Pizza	20%	16
	Curry	30%	24
	Roast	25%	20
	Stir Fry	15%	12
Total		100%	80



Find the Totals: Favourite Colours

Gemma has made a pie chart to show the percentages of children who like different colours. She asked 60 children.

Children's Favourite Colours

Key:

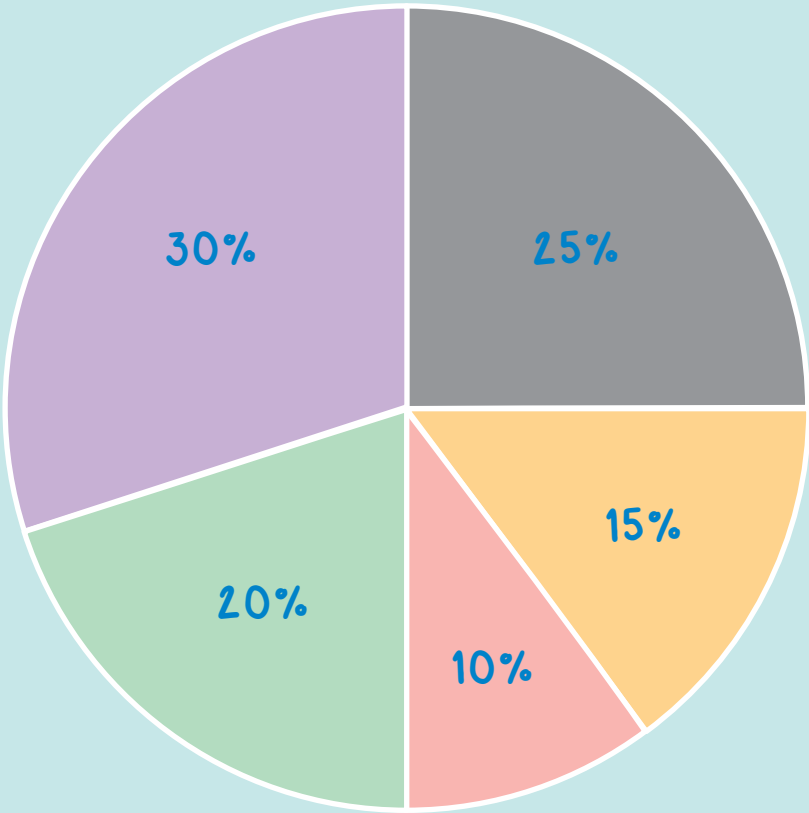
Black

Red

Yellow

Purple

Green



Calculate the totals and complete the table.

Colour	Category	Percentage	Number of Children
	Black		
	Red		
	Yellow		
	Purple		
	Green		
Total		100%	60

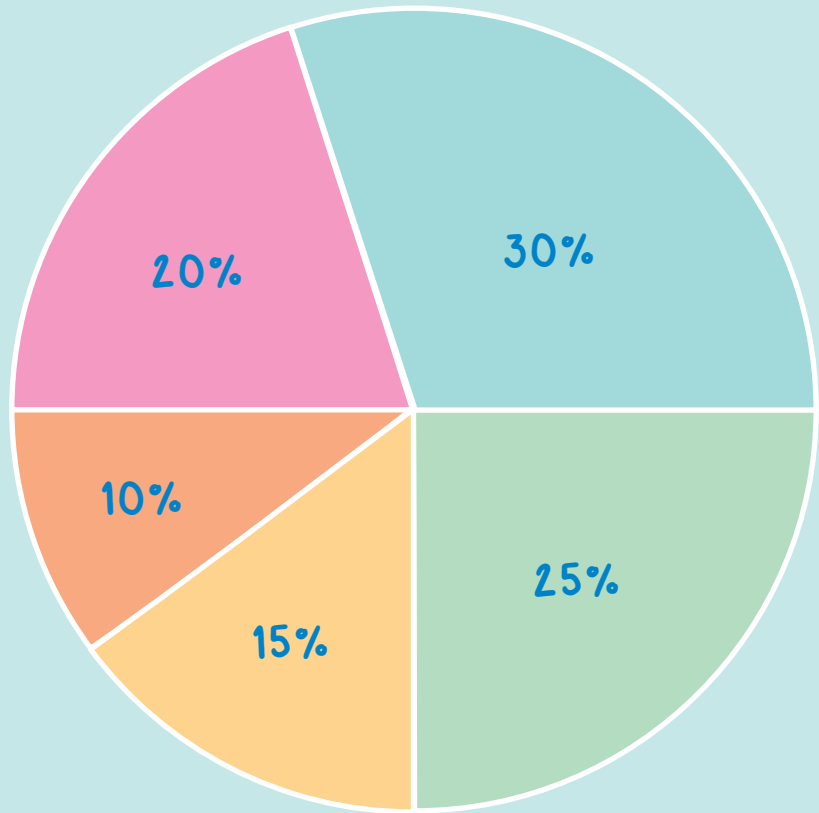
Jayden's Sweet Shop

Jayden sells 5 flavours of fizzy sweets. He sold 240 sweets in one morning. The pie chart shows the percentages of sweets sold during the morning:

Fizzy Sweets Sold

Key:

-  Fizzy Snakes
-  Fizzy Hearts
-  Fizzy Sours
-  Fizzy Apples
-  Fizzy Fish



Use the data to answer the questions.

1. How many fizzy apples were sold?

2. Which was the most popular sweet that morning?

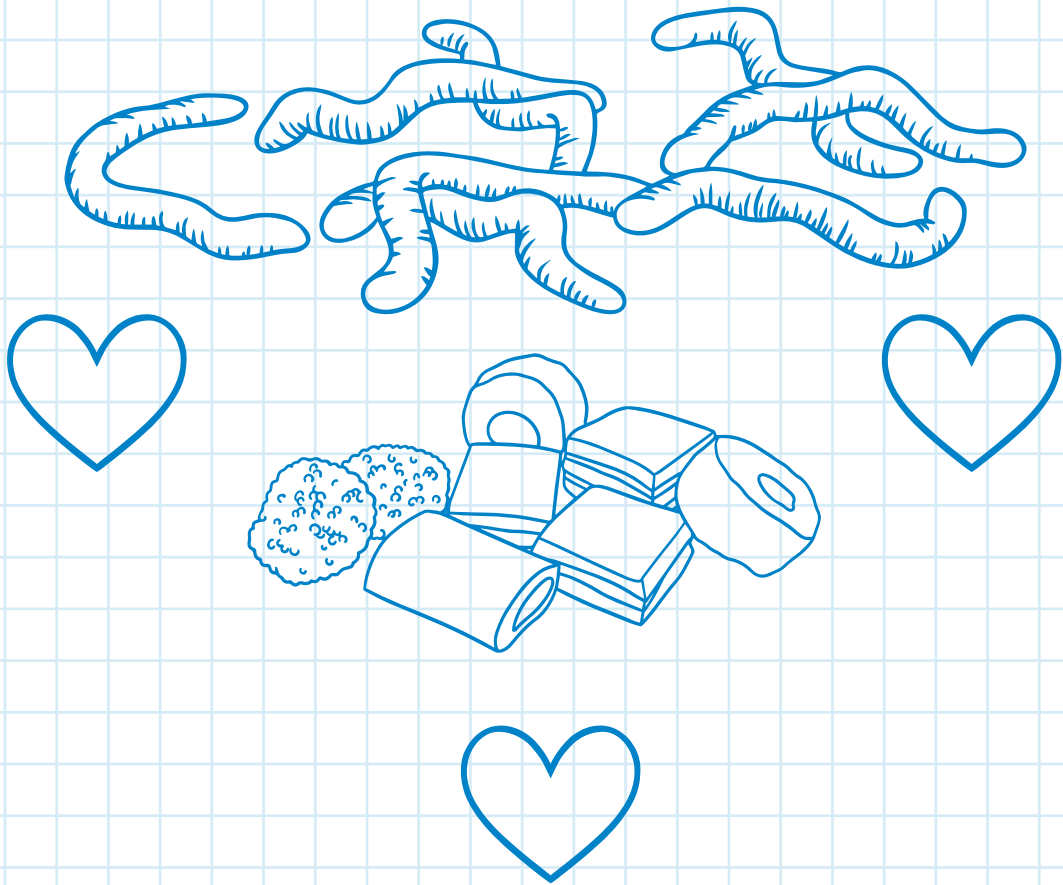
3. Jayden says, "The total percentage of 3 types of sweet was 60%." Which 3 types of sweet could he be talking about?

_____, _____ and _____

4. Mirriam went into the shop that morning and bought 15 fizzy hearts. How many more fizzy hearts were sold that morning?

5. Jayden says, “I sold more fizzy hearts and fizzy apples than fizzy fish and fizzy snakes.”
Is this statement true or false? How can you tell?
6. Use the pie chart to complete the table.

Colour	Category	Percentage	Number of Children
	Fizzy Snakes		
	Fizzy Hearts		
	Fizzy Sours		
	Fizzy Apples		
	Fizzy Fish		
Total		100%	240



Answering Questions from Blank Pie Charts

Sometimes, a question might ask about the quantities in a pie chart that has no numerical data on it. Here is an example:

Hayda Crump is a famous film star. She has made three new films. Arlo asked the members of his class which one was their favourite film. There are 30 people in his class.

Favourite Hayda Crump Films



We can find out the value of the segments because we know that there were 30 people altogether.

The pie chart has been split into 6 segments so one segment represents $30 \div 6 = 5$.

3 segments represent the film Scuba Kings, which means that 3×5 children voted for this film.

$3 \times 5 = 15$ votes for Scuba Kings

1 segment represents the film Magpie Luck, which means that 1×5 children voted for Magpie Luck.

2 segments represent The Missing Sarcophagus film, which means that 2×5 children voted for this one.

$2 \times 5 = 10$ votes for The Missing Sarcophagus

We can check our answers by adding them up – the total should be 30:

$15 + 5 + 10 = 30$

Remember to find out how much one segment is worth first, then you can work out the rest.

Talea's T-Shirts

Talea has been decorating some T-shirts to sell in her shop using different methods. Here is a pie chart to show the methods she used:

T-Shirt Decorations

Key:



Tie Dye



Sequins



Fabric Paint



Graffiti



Talea decorated 78 T-shirts in total.

1. How many tie dye T-shirts did Talea decorate?

2. How many T-shirts have sequins or fabric paint on them?

3. Fill in the gaps with $>$, $<$ or $=$ to show the relationship between the amounts.

tie dye

sequins

tie dye

fabric paint + graffiti

sequins

graffiti

graffiti + sequins

fabric paint

4. How many more T-shirts have fabric paint or tie dye decorations than graffiti?

Bibi's Burger Van

Bibi has been busy with her burger van at the summer fayre. The pie chart below shows the burgers she sold:

Burgers Sold

Key:



Avocado



Halloumi



Chilli



Pulled Pork



In total, Bibi sold 320 burgers.

1. How many halloumi burgers were sold?

2. What is the difference between the most popular and least popular burger?

3. How many burgers sold were chilli or halloumi ones?

4. How many more avocado burgers were sold than pulled pork burgers?

5. Bibi says, "40% of the burgers I sold were halloumi or pulled pork." Is she correct? Explain your answer.

Statistics: Read and Interpret Pie Charts Answers

Favourite Colours Answers

Colour	Category	Percentage	Number of Children
	Black	25%	15
	Red	10%	6
	Yellow	15%	9
	Purple	30%	18
	Green	20%	12
Total		100%	60

Jayden's Sweet Shop Answers

1. How many fizzy apples were sold?

60

2. Which was the most popular sweet that morning?

fizzy snakes

3. Jayden says, "The total percentage of 3 types of sweet was 60%."
Which 3 types of sweet could he be talking about?

fizzy snakes, fizzy hearts and fizzy sours
or

fizzy apples, fizzy fish and fizzy hearts

4. Mirriam went into the shop that morning and bought 15 fizzy hearts.
How many more fizzy hearts were sold that morning?

$$48 - 15 = 33$$

5. Jayden says, "I sold more fizzy hearts and fizzy apples than fizzy fish and fizzy snakes."
Is this statement true or false? How can you tell?

**This statement is incorrect. The amount of fizzy hearts and fizzy apples is 108.
The amount of fizzy fish and fizzy snakes is also 108.**

6. Use the pie chart to complete the table.

Colour	Category	Percentage	Number of Children
	Fizzy Snakes	30%	72
	Fizzy Hearts	20%	48
	Fizzy Sours	10%	24
	Fizzy Apples	25%	60
	Fizzy Fish	15%	36
Total		100%	240

Talea's T-Shirts Answers

1. How many tie dye T-shirts did Talea decorate?

$$13 \times 2 = 26$$

2. How many T-shirts have sequins or fabric paint on them?

$$13 + 26 = 39$$

3. Fill in the gaps with $>$, $<$ or $=$ to show the relationship between the amounts.

tie dye $>$ sequins

tie dye $<$ fabric paint + graffiti

sequins $=$ graffiti

graffiti + sequins $=$ fabric paint

4. How many more T-shirts have fabric paint or tie dye decorations than graffiti?

$$\text{fabric paint} + \text{tie dye} = 26 + 26 = 52$$

$$\text{graffiti} = 13$$

$$52 - 13 = 39$$

Bibi's Burger Van Answers

1. How many halloumi burgers were sold?

$$40 \times 2 = 80$$

2. What is the difference between the most popular and least popular burger?

$$120 - 40 = 80$$

3. How many burgers sold were chilli or halloumi ones?

$$40 + 80 = 120$$

4. How many more avocado burgers were sold than pulled pork burgers?

$$120 - 80 = 40$$

5. Bibi says, "40% of the burgers I sold were halloumi or pulled pork."

Is she correct? Explain your answer.

From the pie chart, we can see that $\frac{2}{8}$ of the burgers were halloumi and $\frac{2}{8}$ were pulled pork. This makes $\frac{4}{8}$, which is the same as $\frac{1}{2}$. $\frac{1}{2}$ as a percentage is 50%, so Bibi is incorrect.